De La Salle Holy Cross College

NOVEMBER EXAMINATION

2024

Grade 10

SPORT AND EXERCISE SCIENCE

Examiner:	Mr S Moore	**Time:**	2 hours
Moderator:	Mrs D Webb	**Marks:**	160

Name:

PLEASE READ THROUGH THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 29 pages and a separate Source Material booklet of 4 pages. Please check that your question paper is complete.
2. **Answer all the questions on the question paper** and hand it in at the end of the examination.
3. Use the total marks awarded for each question as an indication of the detail required.
4. It is in your own interest to write legibly and to present your work neatly.
5. FOUR blank pages (pages 26 – 29) are included at the end of the question paper. If you run out of space for a question, use these pages. Clearly indicate the number of your answer should you use this extra space.

	Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	
Mark									
Total	**10**	**20**	**30**	**20**	**22**	**20**	**18**	**20**	**160**

Remember that God is with you and will help and inspire you in all that you do

QUESTION 1

Write the correct answer in column B to match the correct description in column A.

COLUMN A	COLUMN B
1.1 This type of muscle is controlled voluntarily.	
1.2 This muscle is named after the Greek letter *delta*.	
1.3 This tissue connects muscle to bones.	
1.4 The term *Maximus* means the muscle is ...	
1.5 This muscle is the antagonist for knee flexion.	
1.6 This muscle is the antagonist for elbow flexion.	
1.7 This joint movement is when a limb moves sideways to the body …	
1.8 The term *Inferior* refers to …	
1.9 The term *Supine* means …	
1.10 Abduction occurs in this body plane.	

[10]

QUESTION 2

Examine Picture A and answer the questions that follow.

Picture A



www.skeletal.com

2.1 Identify one bone on which the Olympic Bar is resting on.

_____(1)

2.2 List three bones in the upper limb.

_____

_____(3)

2.3 Name two minerals that are stored in bones.

_____

_____(2)

2.4 Name the joint movement occurring at the knee in Picture A.

_____(2)

2.5 Based on the answer in 2.4 name the muscles acting as the agonist and antagonist for that joint movement.

_____

_____

_____(4)

2.6 If the weightlifter remains in that position for 1 minute, name the type of muscle contraction that is occurring and describe what is happening to the muscle.

_____

_____

_____(3)

2.7 List 5 physiological adaptations that will occur to the **Musculo-Skeletal System only** as a result of doing the exercise in Picture A for a prolonged period of time.

_____

_____

_____(5)

[20]

QUESTION 3

Examine Source A and answer the questions that follow.

Source A

How Energy Systems Training Can Help You Be Better at Your Sport

When athletes and coaches think about conditioning, they often picture endless bouts of Suicide Sprints, Stair Runs and other running drills performed with reckless abandon. Showing up to practice on day one and performing sprints for hours is almost like a rite of passage.

This way of thinking will only get you so far. Better to understand the energy systems in your body and how to train and best use them for your sport.

Many people still believe the aerobic system is only for marathon runners. However, you need to have an aerobic base and build upon it.

The best way to think about your energy systems is to picture a house. You have your basement/foundation, your first floor with the kitchen, bedrooms, and living room and finally your attic. Without a good foundation, the house will eventually crumble. If all you do is high-intensity exercise, you will eventually run out of gas.

Developing a foundation helps us develop more aerobic enzymes, left ventricle eccentric hypertrophy (increased stroke volume and decreased resting heart rate) and mitochondria, among other advantageous adaptations. Our aerobic system allows us to compete for along duration, recover between high-intensity work bouts, buffer our muscles for energy production and maintain power output. Without it, we would all be really great for 15-60 seconds and eventually fall apart.



https://news.umich.edu/

3.1 With reference to the house analogy in Source A, provide the names of the related energy systems.

Basement / Foundation _____

First Floor _____

Attic _____(6)

3.2 Name the organelle from which ATP is produced.

_____(1)

3.3 Provide definitions for Aerobic and Anaerobic energy system.

Aerobic: _____

_____(2)

Anaerobic: _____

_____(2)

3.4 Complete the empty blocks below to show a comprehensive view of energy systems in relation to sporting activities. (18)

Energy System	Fuel used	Duration	Sporting Example
ATP/PC system			
		15 secs – 3 min depending on intensity.	
	Carbohydrates		

3.5 What percentage (range) of one's maximum heart rate could one expect to be in when exercising aerobically?

_____(1)

[30]

QUESTION 4

Examine Source B and answer the questions that follow.

Source B

Is Pull the right term for our arms in Swimming?

I was recently asked if using the term pull for the arm motion in Swimming is the right term. Another term that was suggested for the arm motion was the word leverage. Here is why I disagree with using the term leverage for a swimmer's arm motion.

The word "leverage" implies the use of a lever. In most cases, we think of levers as something that increases the mechanical advantage - enabling us to use a small force to lift or move big objects, but they don't always do that.

The arm is used as a lever in Swimming. In fact, it is a Class III lever. That means that the effort load (muscles moving the arm and hand through the water) and the resistance load (mainly applied to the swimmer's hand) are located on the same side of the lever's fulcrum (shoulder joint).

With the arm, the distance between the fulcrum (shoulder joint) and the effort load (insertion of the muscles) is called the effort arm. The distance is very short, perhaps an inch or two, as the muscles moving the upper arm insert into the head of the humerus bone. The distance from the fulcrum (shoulder) to the resistance load, or hand, is called the load arm. That distance is much longer- more than 30 inches in adults.

https://www.healthline.com/

4.1 Provide a definition of a **lever**.

_____

_____(3)

4.2 Provide the definition of an “Effort Arm”.

_____

_____(3)

4.3 Provide a definition for a “Load Arm”.

_____

_____

_____(3)

4.4 Identify the Effort, Load and Fulcrum in the two pictures below. Once identified, state the class of lever being used.

Picture B



www.athletics.com

Effort _____

Load _____

Fulcrum _____

Class _____(4)

Picture C



www.nba.com

Effort _____

Load _____

Fulcrum _____

Class_____ (4)

4.5 Match the joint movement in column B to the Anatomical Plane in column A. Write only the letter of your chosen description in the answer table below.

COLUMN A		COLUMN B
4.5.1 Frontal Plane	**A.**	Cervical Flexion
4.5.2 Transverse Plane	**B.**	Hip Abduction
4.5.3 Saggital Plane	**C.**	Thoracic Rotation
	D.	Hip Extension
	E.	Cervical Extension
	F.	Hip External Rotation

ANSWER TABLE

Term	Letter
Frontal Plane	
Transverse Plane	
Saggital Plane	

(3)

[20]

QUESTION 5

Examine Source C and answer the questions that follow.

Source C

Carnivore Diet and Endurance Training
Maximizing Performance Naturally

The Carnivore Diet, characterized by an all-meat regimen, is gaining attention in the athletic community, particularly among those involved in endurance training. At its core, this diet consists predominantly of animal products, omitting traditional sources of carbohydrates such as fruits, vegetables, and grains. Advocates suggest that the high protein content integral to the diet aids in muscle repair and recovery, which could be beneficial for athletes who engage in prolonged physical activity. They argue that by prioritizing protein and fats over carbohydrates, an athlete's body can adapt to efficiently utilize fat as a primary energy source, which could potentially enhance performance during endurance sports.

While the Carnivore Diet presents a paradigm shift from conventional endurance training nutrition, its actual influence on athletic performance is an area of ongoing exploration. Athletes contemplating this diet are encouraged to consider both the potential benefits of high-protein intake for muscle repair and the implications of a limited nutrient profile. It is essential for each athlete to assess the diet's alignment with their individual health needs and performance goals, often consulting with a sports nutritionist to ensure the chosen regimen sustains their rigorous training demands and promotes their overall well-being.

https://discover.texasrealfood.com/

5.1 Provide the definition for the term ***calorie***.

_____

_____

_____

_____(3)

5.2 Provide the correct *calorific value* for the standard amount of energy provided by each of the following:

.

Carbohydrate _____

Fat _____

Protein _____(6)

5.3 Would the carnivore diet be classified as “High GI” or “Low GI”? Justify your answer by defining and explaining the term “GI”.

_____

_____

_____

_____

_____

_____

_____

_____(6)

5.4 When should an athlete consume a High GI meal vs a Low GI meal? Justify your answer.

_____

_____

_____

_____

_____(4)

5.5 List three important functions that fat is responsible for with regard to the nervous system.

_____

_____

_____

_____(3)

[22]

QUESTION 6

Examine Source D and answer the questions that follow.

Source D

The Science Of Lance Armstrong

I have heard a lot of comparisons between Michael Phelps and Lance Armstrong. I must admit I don't know too much about swimming, but I do know about cycling, particularly Armstrong. Here are some of the things that I know. The main reason for Armstrong's dominance in his sport is something most great endurance athletes have in common. Yet when it comes to Lance's own individual physiology, he is just nearly superhuman. Here's why:

His genetics for an endurance athlete are nothing short of astounding. Do you know his VO2 Max measures around 83-85 ml/kg/min, while an average person's is around 40? For Lance, his success is attributed to a combination of great training, will, desire, and unbelievable genetic makeup.

He also has a very large heart, his over sized heart can beat over 200 times a minute and thus pump an extraordinarily large volume of blood and oxygen to his legs. His VO2 max—the maximum amount of oxygen his lungs can take in, an important measurement for an endurance athlete is extremely high, some of the highest ever recorded. His heart can pump nine litres of blood per minute (at rest), compared to five for the average person. This number increases to 40L at maximum intensity. His resting heart rate is only 32!

Even though his heart and lungs are something extremely rare he also produces half as much lactic acid and expel it faster, allowing him to ride harder up the Tour's notoriously brutal hills and then recover quickly. With that an average person turns 20 per cent of the oxygen he breathes into muscle power. Armstrong gets 23 per cent - to the highest ever recorded in a human. So with each breath he delivers more power to each push of the pedal.

www.scientificamerican.com

6.1 Using your knowledge of respiratory physiology, explain what V02 Max means.

_____

_____

_____

_____(3)

6.2 List two structures in the respiratory system that will **significantly** influence gaseous exchange and therefore VO2 Max.

_____

_____

_____

_____(2)

6.3 Using your understanding of **energy systems**, explain why V02 Max has a direct effect on endurance performance.

_____

_____

_____

_____

_____

_____

_____(4)

6.4 According to Source D, what is Lance Armstrong's maximum heart rate?

_____

_____(2)

6.5 According to Source D, what is Lance Armstrong's Cardiac Output at rest?

_____

_____(2)

6.6 Calculate Lance Armstrong's **Stroke Volume** when cycling at maximum intensity. Show working.

_____

_____

_____

_____(4)

6.7 List three physiological adaptations that the **respiratory system** would undergo as a result of regular cycling training for 12 months.

_____

_____

_____

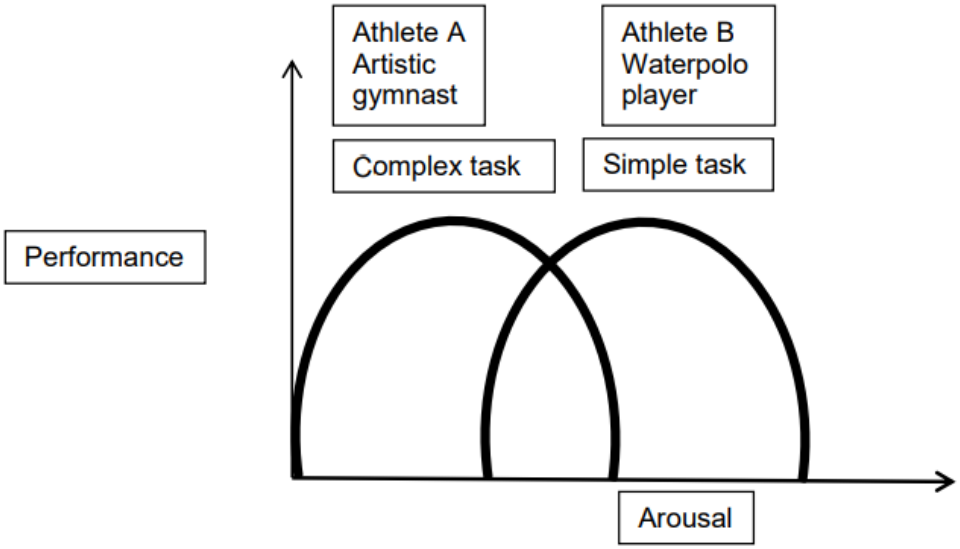
_____(3)

[20]

QUESTION 7

Examine Graph A and answer the questions that follow.

Graph A



https://www.sportpsychology.com/

7.1 Which task on Graph A requires a higher level of arousal?

_____(1)

7.2 Provide justification for your answer in 7.1

_____

_____

_____

_____(3)

7.3 Explain why the two athletes depicted in Graph A might psych themselves up very differently from each other before a major competition.

_____

_____

_____

_____

_____

_____(4)

7.4 Explain the difference in motor skill required to complete a complex task vs a simple task.

_____

_____

_____

_____

_____

_____(4)

7.5 Select two techniques and provide an explanation of how an athlete may decrease their arousal levels, if feeling overwhelmed before a big match or competition.

_____

_____

_____

_____

_____

_____ (4)

7.6 Provide a name for the theory exhibited in Graph A.

_____

_____ (2)

[18]

QUESTION 8

ESSAY

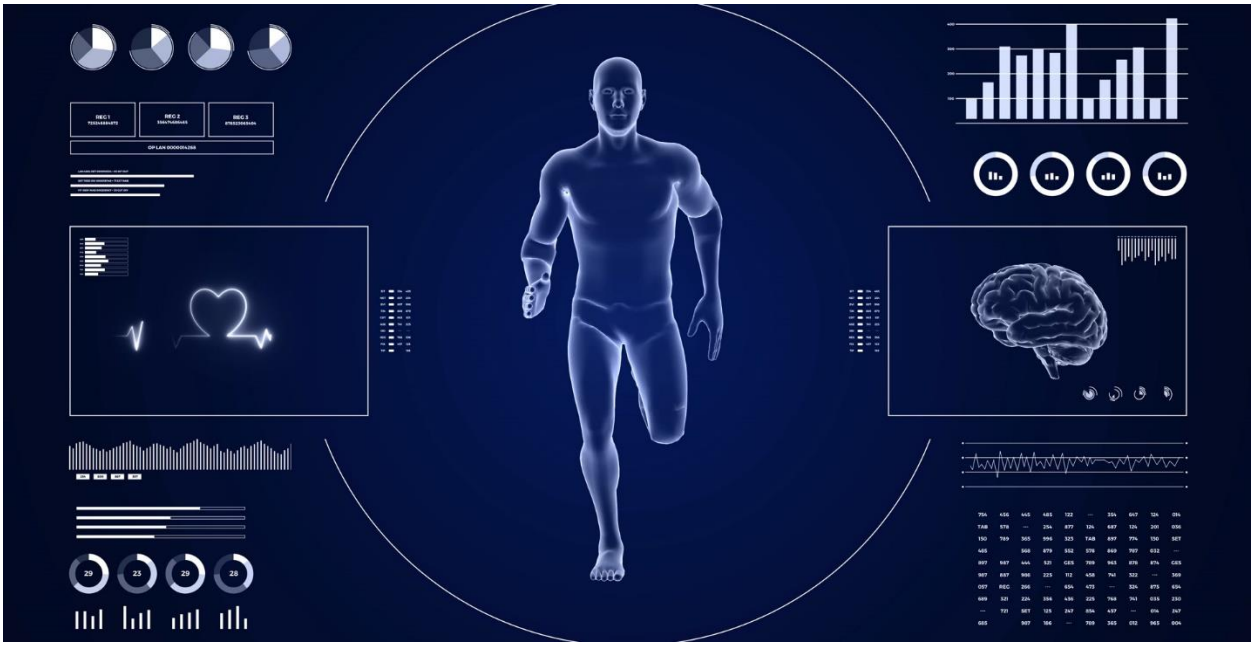
Carefully read and examine the provided sources and the rubric. Use the sources to answer the following question.

Write an essay in which you discuss/argue/debate the following:

Do you think human athletic performance has reached its limit? Provide arguments supporting and challenging this notion.

To answer this question, you are expected to:

- Examine the source material carefully and use the information in the sources to best develop your essay.
- Provide perspectives for and against the notion stated.
- Integrate your own relevant sport science knowledge into your essay.
- Use real-life examples if applicable. **[20]**



https://www.mediclinicinfohub.co.za/

TOTAL 160

ADDITIONAL SPACE (ALL QUESTIONS)

REMEMBER TO CLEARLY INDICATE AT THE QUESTION THAT YOU USED THE ADDITIONAL SPACE TO ENSURE THAT ALL ANSWERS ARE MARKED.